

## Experience

### Microsoft, Developer

Mar 2021 - Present

- Developing AI-Powered applications and embedded cross-device experiences
- Spot Award for excellent work performance in 2022
- Shared Patent Cross-Device Data Transfer Based On A Request-Responding Model (MS#412283-CN-NP) in 2022

### Naturali - a startup, Developer

Aug 2018 - Feb 2021

- Developed a customized version control system for a conversational ChatBots building platform, details in Projects
- Data scraping

### Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Intern

Aug 2017 - Dec 2017

- Visualized the distribution of mobile traffic within commercial areas and uncovered business insights via data mining

### Physical Lab of Northwestern Polytechnical University, Research Assistant

Feb 2017 - Jun 2017

- Created Matlab GUI to support friendly user interaction for the program *Digital Holographic Interferometer for Acquiring 3D Dynamic Information*

### New Oriental Education & Training, Teaching Assistant

Jan 2017 - Mar 2017

- Served as an teaching assistant of English training class in high school section
- Best Creativity Award for my work

## Education

84/100 **B.S in Software Engineering, Northwestern Polytechnical University (NPU) | Xi'an, China**

2014-18

4.14/4.3 **Computer Science and Information Engineering, National Taiwan Normal University | Taipei, Taiwan**

Feb - July 2016

4/5 **Computer Science, University of Oulu | Oulu, Finland**

Feb - Aug 2018

COMPETITIONS(Mathematical Contest in Modeling): The Outstanding Winner of the university in 2016 | The Honorable Mention of Shaanxi Province in 2016

PUBLICATIONS: Lu Jiyang, Panos Kostakos, Mourad Oussalah and Susanna Pirttikangas. *SemanPhone: Combining Semantic and Phonetic Word Association in Verbal Learning Context* 2018 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM). doi:10.1109/ASONAM.2018.8508827 in 2018

## Skills

|                     |                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>  | C#, Golang, Python, C/C++, Kotlin, TypeScript/JavaScript, Java, Racket, HTML+CSS, Bash, Matlab, LaTeX, SQL     |
| <b>Algorithm</b>    | Sorting(Merge Quick Heap), Binary Search, DFS, BFS, Dynamic Programming, Graphing(Dijkstra's Bellman-Ford)     |
| <b>Tools</b>        | Git, Vim, CLI tools, Visual Studio, Android Studio, IntelliJ IDEA, Docker, Postman, Swagger, Wireshark, JMeter |
| <b>Technologies</b> | RESTful, Websocket, SignalR, WebRTC, gRPC, ASP.NET, Gin, Flask, React, Vue, MySQL, PostgreSQL, MongoDB, Azure  |

## Projects

### Build Sharing Feature in Microsoft Copilot

2025

Microsoft

- Goal: Enable a reusable sharing framework in Microsoft Copilot.
- I co-designed and co-implemented a modular sharing framework that allows multiple features to integrate consistent sharing. Migrated existing implementations to the new framework, ensuring maintainability and scalability. Partnered with other teams to onboard their features onto the framework and provided ongoing support. Additionally, I improved the success rate of customized social preview generation for conversation sharing by 20%, enhancing user engagement and share quality.

### Build Quiz Card in Microsoft Copilot

2025

Microsoft

- Goal: Enable an interactive quiz card in conversations within Microsoft Copilot.
- I led the design and evaluation of the core prompt driving quiz card generation, a key component of the feature, including LaTeX support that added significant complexity. Enhanced the quiz card parsing logic, reducing the error rate by 4%.

### Embed Generative AI Capabilities into Android OS

2023 - 2024

Microsoft

- Goal: Integrate generative AI capabilities into the Android operating system.
- I explored AI capability options by evaluating frameworks and building quick prototypes to guide decision-making. I developed phone plugins for managing contact information, created a testing tool to streamline validation. Through collaboration with the Bing Chat Backend team, my work on citation utilities and dynamic prompt leakage filtration was integrated into their system, which powers Bing Chat and co-pilot applications used by millions daily.

### Messages in Microsoft Teams

2022

Microsoft

- Goal: Enable Microsoft Teams users to send and receive SMS messages on their Android phones.
- I developed a push API that allowed phones to notify the Teams app about incoming SMS messages. This involved collaborating with various teams that owned different components. Additionally, to address delays caused by partner teams, I temporarily simulated unfinished components which helped expedite feature testing.

## Cross-Device Data Transfer

2021 - 2022

Microsoft

- Goal: Provide a set of RESTful-style APIs that can be used to fetch device data from anywhere.
- I played a major role in developing a crucial transport service for a microservices architecture. As part of this, I revised a transportation protocol between the transport service and LinkToWindows on phones. I analyzed new requirements for the LinkToWindows side and communicated with the cross-geographical team to ensure success. I implemented some layers and features of the protocol, such as continuous chunk responses, shared connection management. To enhance the service's concurrency, I delved into advanced C# asynchronous programming techniques. Lastly, I ensured good debuggability by defining reasonable and helpful telemetry events.

## Conversational User Interface for Building Conversation ChatBot

2019 - 2020

Naturali

- Goal: Create a platform and set of APIs that customer developers can use to build ChatBots, and create a ChatBot community similar to GitHub.
- As a developer for the ChatBot building platform, I created a version control system in the backend service to enhance collaboration among customer developers. To enable submission of pull requests to the main ChatBot version for reviewing changes, I implemented differentiating and merging versions. To facilitate the importation of ChatBot components from public libraries, I implemented referencing or forking a ChatBot component. To accommodate the creation of template ChatBots for various languages, I implemented partitioning and consolidating language-specific and non-language-specific aspects of ChatBot development. I made an informed decision on a fundamental storage strategy that ultimately proved superior in both development simplicity and overall performance.

## Data Scraping

2018

Naturali

- Goal: Build automatic pipeline to collect large volume various types of data for model training purposes.
- Initially, I determined the data sources to gather based on the data requirements, e.g. QQ Music. Next, I analyzed the XHR requests made to the data source and customized a Python web crawling script to retrieve information. Also I maintained and improved performance of the overall scraping framework continuously.